Module/Pathway/Taxon Review: Bacterial Pst High-Affinity Phosphate Uptake in *Pseudomonas putida* KT2440

Target taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Target bucket:** KEGG ppu02010 "ABC transporters" (151 primary genes; 207 candidates; module area: transport_motility_signaling) **Module reviewed:** Bacterial Pst high-affinity phosphate uptake (PstS capture → PstA/PstC translocation → PstB ATP coupling)

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1. Executive Summary

The bacterial Pst high-affinity phosphate-uptake module is **fully satisfiable (COVERED)** in *Pseudomonas putida* KT2440. All three canonical module steps — periplasmic phosphate capture, phosphate-selective membrane translocation, and ATP-dependent energy coupling — are encoded by a single, contiguous Pho-regulon operon on the KT2440 chromosome: **PP_5329 (pstS)**, **PP_5328 (pstC)**, **PP_5327 (pstA)**, and **PP_5326 (pstB)**, immediately followed by **PP_5325 (phoU)** and adjacent to the **PhoB/PhoR** two-component regulatory system (PP_5320/PP_5321). This architecture — *pstSCAB-phoU* beside *phoBR* — is the canonical bacterial Pho regulon layout, and KEGG correctly partitions the transport genes into ppu02010 (ABC transporters) while placing the regulatory genes in ppu02020 (two-component system).

The module assignment is supported by **direct, same-species functional evidence**. Wu et al. (1999; PMID: 16232467) cloned the *pstSCAB* genes of *Pseudomonas putida* PRS2000 and showed that chromosomal inactivation of *pstSC* abolished ^{32}Pi uptake even under phosphate limitation, while causing constitutive alkaline phosphatase synthesis and Pi chemotaxis. This demonstrates that PstSCAB is both the high-affinity Pi transporter and a negative regulator of the *pho* regulon in this species. Because KT2440's Pho-box/PhoBR-adjacent operon is orthologous to the PRS2000 operon (and the encoded proteins share >75% identity even with *P. aeruginosa* PAO1), the functional transfer to KT2440 is **strong**.

For curators, the principal issues are **not gaps but paralog ambiguity and over-annotation** within the broad ppu02010 bucket. KT2440 carries (i) a second, complete but **non-Pho-linked** Pst-type operon (PP_2656–2659) whose physiological role is uncertain; (ii) a standalone **OmpA–phosphate-binding fusion** (PP_3818) that is likely mis-annotated as a classic PstS; and (iii) a distinct **phosphonate (C–P bond) ABC transporter** (PhnCE/PtxABCD, PP_0824–0827) plus **low-affinity non-ABC Pi carriers** (PiT-family PP_4103, Na^+/Pi symporter PP_1373). These belong to separate modules and should be kept out of the inorganic-Pi Pst module. The recommended curation outcome is: **all three Pst steps** = **covered** via PP_5326–5329; PP_2656–2659 = **candidate_uncertain**; PP_3818, PP_0824–0827, PP_4103, PP_1373 = **excluded from this module** (belong to adjacent modules).

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2. Target-Organism Pathway Definition

2.1 What the module includes

The Pst (phosphate-specific transport) module is the **ATP-binding-cassette (ABC) importer for inorganic orthophosphate (Pi)** operating under phosphate-limiting conditions. It comprises exactly three biochemical steps:

1. **Periplasmic phosphate capture** — the substrate-binding protein PstS scavenges free Pi in the periplasm with high affinity.
2. **Phosphate-selective membrane translocation** — the transmembrane permease pair PstA/PstC forms the channel through which Pi passes across the inner membrane.
3. **ATP-dependent energy coupling** — the cytoplasmic ATPase PstB (EC 7.3.2.1) hydrolyzes ATP to drive Pi import against its concentration gradient.

The two mechanistic relationships defined in the module scope both hold in KT2440: phosphate-loaded PstS presents substrate to the PstA/PstC permease, and PstB ATP hydrolysis drives phosphate passage through PstA/PstC.

2.2 What must be kept separate

- **PhoU-mediated homeostasis (PP_5325)** and the **PhoR/PhoB starvation response (PP_5321/PP_5320)** are *adjacent regulatory systems*, not transport steps. They are genomically part of the same locus but are correctly mapped by KEGG to ppu02020 (two-component system), and should remain outside the transport module.

- **Phosphonate uptake (PhnCE / PtxABCD; PP_0824–0827)** is a chemically distinct system that imports organophosphonates (C–P bond compounds), not inorganic orthophosphate. It is a separate Pho-regulon member (KO K02041/K02042/K02044; EC 7.3.2.2).
- **Low-affinity Pi uptake** is provided by non-ABC carriers — the PiT-family symporter **PP_4103** (KO K03306) and a phosphate:Na^+ symporter **PP_1373** (KO K16322) — which are not ABC transporters and belong to different transport modules.
- Broad **overview maps** (KEGG "ABC transporters" ppu02010 as a whole) mix ~207 candidate genes covering dozens of unrelated substrates (amino acids, metals, sugars, LPS export, etc.). Only the Pst-specific loci belong to this module.

2.3 Alternate names and database definitions

- **Pst** = "phosphate-specific transport" system; genes *pstS*, *pstC*, *pstA*, *pstB* (sometimes *pstB1/pstB2* for paralogs).
- KEGG orthologs: **K02040 (pstS), K02037 (pstC), K02038 (pstA), K02036 (pstB)**; PhoU = K02039.
- EC number for the ATPase: **7.3.2.1** (ABC-type phosphate transporter). Note the naming inversion in UniProt: the ATPase of the canonical Pho-linked operon (PP_5326) is annotated **pstB2**, while the paralog ATPase (PP_2659) is **pstB1** — an inversion relative to genomic/regulatory context that curators should flag.

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3. Expected Step Model and Satisfiability

Module step	Expected player	KT2440 gene(s)	KEGG KO	Status
1. Periplasmic phosphate capture	PstS (phosphate-binding protein)	**PP_5329** (pstS)	K02040	**Covered**
2. Phosphate-selective translocation	PstA/PstC permease pair	**PP_5327** (pstA) + **PP_5328** (pstC)	K02038 / K02037	**Covered**
3. ATP-dependent energy coupling	PstB ABC ATPase (EC 7.3.2.1)	**PP_5326** (pstB2)	K02036	**Covered**
(Adjacent) Homeostasis regulator	PhoU	PP_5325	K02039	Out of module (ppu02020)
(Adjacent) Starvation two-component system	PhoR/PhoB	PP_5321 / PP_5320	K07636 / K07657	Out of module (ppu02020)

All three module steps are encoded by candidate genes in the queried ppu02010 bucket. No step is missing. The Pst genes fall in the high-PP-number range (PP_5326–5329) and were therefore among the 127 candidate genes truncated from the prompt excerpt (which cut off at PP_1139), not absent from the metadata — this was explicitly verified against `KEGG link/ppu/path:ppu02010` (13/13 relevant loci present, 0 absent).

No Pst step is "not expected" in this organism; *P. putida* is a soil bacterium that routinely experiences phosphate limitation and possesses the full canonical Pho regulon.

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4. Candidate Genes and Evidence

4.1 Finding F001 — The canonical pstSCAB-phoU operon (PP_5325–PP_5329)

KEGG (ppu) and UniProt (proteome UP000000556, taxon 160488) jointly identify a contiguous, single-strand (complement) gene cluster on the KT2440 chromosome that encodes all three module steps:

- **PP_5329 pstS** — phosphate-binding protein, periplasmic capture (UniProt Q88C54, 332 aa)
- **PP_5328 pstC** + **PP_5327 pstA** — permease pair, membrane translocation (Q88C55 / Q88C56)

- **PP_5326 pstB2** — phosphate import ATP-binding protein, EC 7.3.2.1 (Q88C57, 277 aa)

Immediately downstream lies **PP_5325 phoU** (PhoU accessory regulator; Q88C58) and the **PhoB/PhoR two-component system** (PP_5320/PP_5321; Q88C63/Q88C62). Genomic positions (~6.070–6.078 Mb, all complement strand) are consistent with a single *pstSCAB-phoU* operon adjacent to *phoBR* — the canonical bacterial Pho regulon architecture. KEGG maps PP_5326–5329 to ppu02010 (ABC transporters) while *phoU/phoB/phoR* map to ppu02020 (two-component system), correctly excluded from the transporter map. **This is the high-confidence, module-satisfying operon.**

4.2 Finding F003 — Direct same-species functional validation

Wu et al. (1999; PMID: 16232467) cloned the *pstSCAB* genes of *Pseudomonas putida* PRS2000. Predicted products showed **83 / 75 / 78 / 88 % amino-acid identity** (PstS / PstC / PstA / PstB) to the *P. aeruginosa* PAO1 counterparts. Two Pho box sequences were found upstream of *pstS* (15/18 base identity to consensus) and in the *pstS–pstC* intercistronic region (11/18). Chromosomal inactivation of *pstSC* (mutant PNT1) **abolished ^{32}Pi uptake even under Pi limitation** and caused constitutive alkaline phosphatase synthesis and Pi chemotaxis, showing PstSCAB is both the high-affinity Pi transporter and a negative regulator of the *pho* regulon.

> *"The resultant mutant, designated PNT1, failed to take up ^{32}Pi even under conditions of Pi limitation."* — PMID: 16232467
>
> *"Two well-conserved Pho box sequences were found in the region upstream of the pstS gene."* — PMID: 16232467

UniProt/KEGG place the orthologous, Pho-box/PhoBR-adjacent operon in KT2440 at PP_5329/PP_5328/PP_5327/PP_5326 (Q88C54/55/56/57). **Species transfer is STRONG** (same species, different strain; >75% identity even to *P. aeruginosa*).

4.3 Findings F002 & F004 — Paralog landscape and domain architecture

KEGG KO mapping returns multiple ppu genes per phosphate KO, revealing paralog ambiguity:

KO	Gene	Canonical (Pho-linked)	Paralog	Third copy
K02040 pstS	phosphate-binding	PP_5329	PP_2656	PP_3818
K02037 pstC	permease	PP_5328	PP_2657	—
K02038 pstA	permease	PP_5327	PP_2658	—
K02036 pstB	ATPase	PP_5326 (pstB2)	PP_2659 (pstB1)	—

Cluster 2 (PP_2656–2659) is a second complete *pst* operon at ~3.043–3.047 Mb (forward strand). Domain/transmembrane analysis clarifies the roles:

- **Phosphate-binding subunits:** PP_5329 (PF12849, 332 aa) and PP_2656 (PF12849, 348 aa) are classic PBP-type PstS. **PP_3818 (435 aa) carries PF00691 (OmpA) fused to PF12849** — an outer-membrane-associated phosphate-binding fusion, *not* a canonical periplasmic PstS.
- **Permeases:** all PF00528 (BPD_transp_1). Canonical **PP_5327 (556 aa, 6 TM)** and **PP_5328 (762 aa, 8 TM)** are genuine ABC permeases but markedly **longer** than the paralogous PP_2658 (297 aa, 6 TM) and PP_2657 (322 aa, 7 TM), which match canonical PstA/PstC sizes (~300–320 aa).
- **ATPases:** PP_5326 and PP_2659 are both PF00005 (ABC_tran, ~277–279 aa).

All these entries are UniProt evidence level "Inferred from homology" or "Predicted."

4.4 Finding F006 — Genomic-context distinction of the paralog

The second Pst-type operon PP_2656–2659 is flanked by PP_2653 (Cro/CI-family transcriptional regulator), PP_2654 (glutathione S-transferase), and hypothetical proteins PP_2655/PP_2660/PP_2661 — i.e., **NOT adjacent to phoU/phoB/phoR**, unlike the canonical PP_5325–5329 / PP_5320–5321 locus. This absence of Pho-regulon context is the key argument that PP_2656–2659 is *not* the primary high-affinity Pi transporter and should be treated as candidate_uncertain (possibly serving a specialized or conditionally expressed role).

PP_3818 (OmpA + phosphate-binding fusion) is isolated among a polyamine ABC transporter (PP_3816/PP_3817), glutathione reductase (PP_3819), and a group II intron maturase (PP_3820), **with no cognate permease/ATPase neighbours** — reinforcing that it is not a functional Pst substrate-binding subunit.

4.5 Finding F005 — Phosphonate transporters are a distinct system

The ppu02010 candidate list includes **PP_0824 ptxB / PP_0825 phnC (EC 7.3.2.2) / PP_0826 phnE / PP_0827 ptxC** — a phosphonate (C–P bond) uptake ABC transporter (KO K02041/K02042/K02044), biochemically distinct from inorganic-phosphate PstSCAB (KO K02036–K02040, EC 7.3.2.1). Comparative *Pseudomonas* Pho-regulon proteomics (Lidbury et al. 2016, PMID: 27233093) found putative phosphonate transporters among Pi-depletion-induced PHO proteins, and *P. putida* can degrade organophosphonates such as 2-aminoethylphosphonate (PMID: 9841125). Both PstSCAB and PhnCE are Pho-regulon members but serve **different substrates**.

> *"...several proteins, previously not associated with the response to Pi depletion, were also identified. These included putative nucleases, phosphotriesterases, putative phosphonate transporters and outer membrane proteins."* — PMID: 27233093

4.6 Finding F007 — Coverage is real, not a metadata gap

`KEGG link/ppu/path:ppu02010` confirms all relevant loci are members of the queried bucket: the PstS/C/A/B primary operon (PP_5326, PP_5327, PP_5328, PP_5329); the paralog operon (PP_2656, PP_2657, PP_2658, PP_2659); the OmpA-fusion PP_3818; and the phosphonate transporter PP_0824–PP_0827 (**13/13 present; 0 absent**). These loci fall in the high-PP-number range and were among the candidate genes omitted from the truncated prompt excerpt, not missing from the metadata.

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5. Mechanistic Model / Interpretation

The KT2440 high-affinity phosphate-uptake landscape can be summarized as a single functional Pst module embedded in the Pho regulon, surrounded by paralogs and alternative carriers that must be kept out of the module:

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            PERIPLASM                      INNER MEMBRANE          CYTOPLASM
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Pi   ──▶   PstS (PP_5329) ──presents──▶   PstC (PP_5328) │ PstA (PP_5327)  ──▶ Pi in
           phosphate capture              permease pair (translocation)
           [K02040]                       [K02037 / K02038]
                                                   ▲
                                                   │ ATP hydrolysis drives transport
                                                   │
                                          PstB2 (PP_5326)  [K02036, EC 7.3.2.1]
                                          ATP ──▶ ADP + Pi

Regulatory context (NOT part of transport module, KEGG ppu02020):
   PhoU (PP_5325) ── homeostasis ──┐
   PhoR (PP_5321) ─ sensor kinase ─┼─▶ PhoB (PP_5320) ─▶ Pho-box activation
                                   │      of pstSCAB & pho regulon
   (Pho boxes upstream of pstS)  ──┘
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Parallel / alternative phosphorus systems (separate modules):

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┌──────────────────────────────────────────────────────────────────────┐
│ Second Pst operon  PP_2656–2659  (forward strand, ~3.04 Mb)          │
│   NO phoU/phoBR neighbours → candidate_uncertain                     │
│   canonical-size permeases (297/322 aa), ATPase = pstB1              │
├──────────────────────────────────────────────────────────────────────┤
│ PP_3818  OmpA(PF00691)+PBP(PF12849) fusion, 435 aa, no permease/ATPase│
│   → likely mis-annotated PstS; NOT a canonical binding subunit       │
├──────────────────────────────────────────────────────────────────────┤
│ Phosphonate ABC importer  PhnCE/PtxABCD  PP_0824–0827 (EC 7.3.2.2)   │
│   → C–P bond substrates, separate module                             │
├──────────────────────────────────────────────────────────────────────┤
│ Low-affinity Pi carriers (non-ABC): PiT PP_4103 (K03306),            │
│   Na+/Pi symporter PP_1373 (K16322)  → separate modules              │
└──────────────────────────────────────────────────────────────────────┘
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The interpretive core is that **genomic context is decisive for disambiguating paralogs**. Two features cleanly separate the true module-satisfying operon from its look-alikes: (1) adjacency to *phoU-phoBR* and the presence of upstream Pho boxes (present only for PP_5326–5329), and (2) the concordance of same-species knockout phenotype with this operon's orthology. The paralog operon lacks the regulatory context, and the OmpA fusion lacks both the correct domain architecture and cognate transport partners. This is a textbook case where an unfiltered KEGG ABC-transporter bucket over-reports "phosphate transporter" candidates because homology-based KO assignment cannot by itself distinguish the physiologically dominant transporter from paralogs and chemically distinct systems.

An important curation caveat is the **UniProt pstB1/pstB2 naming inversion**: the ATPase of the canonical Pho-linked operon (PP_5326) is named *pstB2*, while the paralog (PP_2659) is *pstB1*. Curators relying on a "pstB1 = primary" heuristic would pick the wrong locus; the genomic/regulatory evidence should override the name.

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6. Gaps, Ambiguities, and Likely Over-Annotations

Item	KT2440 loci	Issue	Recommended handling
Paralog Pst operon	PP_2656–2659	Complete *pst*-type operon but **no Pho-regulon context**; unknown physiological trigger	**candidate_uncertain** — promote to full review
OmpA–PBP fusion	PP_3818	OmpA(PF00691)+PstS(PF12849) fusion, no permease/ATPase partners; likely **over-propagated "PstS" annotation**	Exclude from Pst module; flag annotation
Phosphonate transporter	PP_0824–0827	Distinct substrate (C–P bond, EC 7.3.2.2); Pho-regulon member but **different module**	Exclude; assign to phosphonate module
Low-affinity Pi carriers	PP_4103 (PiT), PP_1373 (Na^+/Pi)	Non-ABC; **not part of Pst ABC module**	Exclude; separate transporter modules
pstB naming inversion	PP_5326 (pstB2) vs PP_2659 (pstB1)	UniProt names invert genomic/regulatory context	Curator note; trust genomic context
Elongated canonical permeases	PP_5327 (556 aa), PP_5328 (762 aa)	Longer than typical PstA/PstC (~300 aa) but genuine PF00528 permeases	Accept as genuine; note length atypicality
Broad ppu02010 bucket	~207 candidate genes	Overview map mixes dozens of unrelated ABC substrates	Only PP_5326–5329 belong to this module

Key uncertainty: All KT2440 Pst annotations are UniProt evidence level "Inferred from homology" or "Predicted." The functional demonstration is in *P. putida* PRS2000, not KT2440 directly — a same-species but different-strain transfer (strong, but not strain-direct experimental proof for KT2440 itself).

7. Module and GO-Curation Recommendations

7.1 Module step status

- **Step 1 (PstS phosphate capture): COVERED** by PP_5329 (K02040). High confidence.
- **Step 2 (PstA/PstC translocation): COVERED** by PP_5327 + PP_5328 (K02038/K02037). High confidence.
- **Step 3 (PstB ATP coupling): COVERED** by PP_5326 (K02036, EC 7.3.2.1). High confidence.
- **Overall module: COVERED / satisfiable** in *P. putida* KT2440 via the single operon PP_5326–5329.

7.2 Module boundary recommendations

- The **generic module boundaries are correct** for this organism. PhoU/PhoR/PhoB should remain *outside* the transport module as adjacent regulatory systems (KEGG already segregates them to ppu02020).
- **No module_needs_revision** flag for the core three steps.
- **No new GO term requests** appear necessary; existing GO terms cover phosphate ion binding (GO:0042301), phosphate ion transmembrane transporter activity (GO:0015415 / GO:0005315), and ATPase-coupled phosphate transport.

7.3 Genes to exclude / reassign

- PP_2656–2659 → **candidate_uncertain** (do not count toward primary coverage; investigate role).
- PP_3818 → exclude from Pst module; likely **over-annotation** of the PstS role.
- PP_0824–0827 → belongs to a **phosphonate uptake module**, not inorganic-Pi Pst.
- PP_4103, PP_1373 → **low-affinity / non-ABC Pi transport**, separate modules.

8. Genes to Promote to Full `fetch-gene` Review

1. **PP_5329 (pstS), PP_5328 (pstC), PP_5327 (pstA), PP_5326 (pstB2)** — the four module-satisfying genes; promote to confirm strain-direct annotation and lock coverage. *Priority: high (module-defining).*
2. **PP_2656–2659 (second Pst operon)** — resolve whether this is a functional Pi transporter, a specialized/conditional paralog, or a pseudo-operon. Its lack of Pho context and canonical-size permeases make its role the single most important open question. *Priority: high (ambiguity resolution).*
3. **PP_3818 (OmpA–PBP fusion)** — confirm and correct the likely over-propagated "PstS" annotation; determine actual function (possibly outer-membrane phosphate scavenging). *Priority: medium.*
4. **PP_0824–0827 (PhnCE/PtxABCD)** — confirm phosphonate-transport assignment and route to the correct module. *Priority: medium (boundary hygiene).*

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9. Evidence Base and Key References

PMID	Title (abbrev.)	Role in this review	Evidence strength for KT2440
16232467	*Cloning and characterization of P. putida pst system*	**Primary functional evidence**: pstSC knockout abolishes ^{32}Pi uptake; Pho boxes upstream of pstS	Strong (same species, PRS2000 strain)
27233093	*Comparative genomic/ proteomic analyses of three Pseudomonas strains*	Distinguishes phosphonate transporters as a separate Pi-depletion-induced class	Moderate (genus-level)
9841125	*2-aminoethylphosphonic acid biodegradation in P. putida NG2*	Confirms *P. putida* organophosphonate metabolism, supporting phosphonate-system separation	Moderate (species, different strain)

Supporting context papers (phosphate-related *P. putida* physiology, not module-defining): PMID: 41264166 (aluminium exposure perturbs phosphate metabolism in KT2440); PMID: 39522389, PMID: 38360401, PMID: 36966146 (*P. putida* phosphate-solubilization / plant-interaction studies).

Database evidence: KEGG (organism ppu; pathways ppu02010, ppu02020; KO K02036–K02040), UniProt (proteome UP000000556; Q88C54–Q88C58, Q88C62/C63; Pfam PF12849, PF00528, PF00005, PF00691), NCBI taxon 160488.

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10. Limitations and Knowledge Gaps

1. **No strain-direct functional data for KT2440.** The decisive knockout evidence is from *P. putida* PRS2000, not KT2440. Orthology and >75% identity make transfer strong, but a KT2440-specific *pstSC* deletion phenotype has not been cited here.
2. **All KT2440 Pst annotations are homology-inferred or predicted** (UniProt evidence levels), not experimentally curated at the protein level.
3. **The role of the second Pst operon (PP_2656–2659) is unknown.** Without expression or knockout data, we cannot say whether it contributes to Pi uptake, transports a different oxyanion, or is silent.
4. **PP_3818's actual function is unverified** — the OmpA fusion suggests outer-membrane phosphate handling, but this is inferred from domain architecture only.
5. **Pi-affinity kinetics (Km) for KT2440's transporters have not been measured** in the data reviewed; "high-affinity" status rests on the Pho-regulon linkage and PRS2000 phenotype.

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11. Proposed Follow-up Experiments / Actions

1. **Strain-direct genetics:** Construct a KT2440 Δ*pstSCAB* (ΔPP_5326–5329) mutant and measure ^{32}Pi/^{33}Pi uptake and alkaline phosphatase derepression to confirm strain-specific function.
2. **Paralog dissection:** Single and double deletions of PP_2656–2659 vs PP_5326–5329, with growth on limiting Pi and transcriptomics under Pi starvation, to define the paralog's role and Pho-regulon (in)dependence.

3. **Expression profiling:** RNA-seq / proteomics under Pi-replete vs Pi-depleted conditions to test which operon(s) are Pho-regulated (expect PP_5326–5329 induced, PP_2656–2659 possibly not).
4. **PP_3818 characterization:** Localization and phosphate-binding assays to test the outer-membrane phosphate-scavenging hypothesis and correct the annotation.
5. **Curation actions:** Mark the three Pst steps covered; flag PP_2656–2659 candidate_uncertain and promote to full review; reassign PP_0824–0827 to a phosphonate module; add a curator note on the pstB1/pstB2 naming inversion.
6. **Expert question:** Ask a *Pseudomonas* phosphate-physiology expert whether the second Pst operon is known to be functional or conditionally expressed in any *P. putida* strain.

Review complete. Module: Bacterial Pst high-affinity phosphate uptake — COVERED and satisfiable in P. putida KT2440 via PP_5326–5329, with paralog and boundary curation flags delivered.

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